

Final Summary: Evaluating a New Search Ranking System

Phase 1: Data Preparation

Two datasets were prepared for this analysis: a sessions dataset containing booking activity and a users dataset containing each user's assigned experiment group. Both were inspected for structure and missing values before being combined. The two datasets were joined on the user ID so that every session could be matched to the correct experiment group, either control (the existing search system) or variant (the new search system). A primary metric called conversion was then created, marked as 1 when a session ended in a booking and 0 when it did not. This step ensured the data was clean, aligned, and ready for fair comparison between the two groups.

Phase 2: Sanity Check Before Analysis

Before trusting any result, the experiment setup itself was checked for fairness. The concern here is Sample Ratio Mismatch, which happens when users are not split evenly between the two groups as intended. An uneven split can quietly distort results and lead to wrong conclusions.

The split was almost perfectly balanced:

- The control group contained 7,630 users and the variant group contained 7,653 users.
- A chi square test returned a p value of 0.8524, far above the significance level of 0.10.

Because the p value was high, there was no sign of any imbalance. This confirmed the experiment was set up correctly and the results could be trusted.

Phase 3: Measuring the Effect

With the setup validated, the two groups were compared on a primary metric and a guardrail metric. The primary metric measures whether the new system improves bookings. The guardrail metric checks that the new system does not accidentally harm the user experience while doing so.

Primary metric: Conversion

- A test of proportions returned a p value of 0.0002, well below the 0.10 threshold, meaning the result is highly unlikely to be random chance.
- The effect size was positive at 0.1422, meaning the variant lifted conversion by about 14 percent relative to control.

- Together this shows a clear and genuine improvement in bookings under the new search system.

Guardrail metric: Time to booking

- A test on time to booking returned a p value of 0.5365, which is above the threshold and therefore not significant.
- The effect size was a very small negative value of 0.0079, meaning booking time barely changed.
- This shows the new system did not slow users down or harm their experience in any meaningful way.

Overall Insight

The analysis shows that the new search ranking system delivers a real and meaningful improvement in conversion, lifting bookings by roughly 14 percent, while leaving the user experience unharmed. The experiment was confirmed to be fairly run, the primary result was strongly significant, and the guardrail metric stayed safe.

Because the new system improved the main goal without any negative side effects, the final decision is to roll it out fully. This step by step approach, from careful data preparation through validation and statistical testing, demonstrates how a product decision can be grounded in clear evidence rather than intuition.